# Hariharan Babu

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## EDUCATION

**M.S. Data Science** — May 2026
University of North Texas, Dallas, TX — 4.0 GPA

**B.E. Computer Science and Engineering** — May 2024
Anna University College of Engineering, TN, India — 3.5 GPA

## PROFESSIONAL EXPERIENCE

**Data Engineer, Newgen KnowledgeWorks, Full-Time, Chennai, TN, India** — Jun 2023 - Jul 2024

- Architected a serverless, event-driven ETL pipeline on AWS (S3, Lambda, Glue), automating ingestion of 5M+ records/day with 99.9% data availability and reducing manual processing effort by 40%.
- Developed scalable transformation workflows using AWS Glue and DataBrew to process complex nested JSON data, improving data quality by 30% and reducing downstream query latency by 25%.
- Engineered a fault-tolerant and secure framework using SQS, DLQ, AES-256 encryption, OpenSearch, and CloudWatch, decreasing pipeline failures by 35% and ensuring real-time monitoring with proactive alerting.

**Data Science, IHUB Divya Sampark at IIT-Roorkee, Internship** — Jun 2023 - Jul 2023

- Led end-to-end data science workflow (data cleaning, EDA, feature engineering, validation) on real-world datasets, delivering predictive models with 92% accuracy, 0.90 F1-score, and 0.94 ROC-AUC, driving statistically reliable insights.
- Developed and validated machine learning models using Python (Pandas, NumPy, Scikit-learn), improving prediction precision by 25% and reducing error rate by 18%, with robust cross-validation and performance benchmarking.

**Machine Learning using Python, National Institute of Technology (NITK), Internship** — Jan 2023 - Feb 2023

- Developed and optimized supervised ML models (Logistic Regression, Decision Trees, KNN) in Python, increasing prediction accuracy to 91% with 0.89 F1-score and 0.93 ROC-AUC through feature engineering and hyperparameter tuning.
- Built an end-to-end ML pipeline including preprocessing, scaling, and cross-validation, reducing overfitting by 22% and improving model generalization with <5% train–validation performance gap.

**Machine Learning, IEEE, Internship** — May 2022 - Jun 2022

- Implemented supervised and unsupervised ML algorithms (Regression, Classification, K-Means) in Python, achieving 89% classification accuracy and improving convergence speed by 20% through optimized hyperparameter tuning.
- Developed and evaluated ML workflows using Scikit-learn, performing cross-validation and performance benchmarking with 0.87 F1-score and 0.91 ROC-AUC, ensuring robust model generalization.

## TECHNICAL SKILLS

- **Programming**: Python (Pandas, PySpark), SQL (Joins, CTEs, Window Functions, Query Optimization), Java, C/C++, JavaScript.
- **Cloud Platforms:** AWS (S3, EC2, EMR, Glue, Lambda, EKS, IAM, RDS, Redshift, CloudWatch), GCP (BigQuery, Dataflow), Azure (Azure Data Factory, Azure Databricks).
- **Big Data Technologies:** Apache Spark (PySpark), Hadoop (HDFS), Kafka, MapReduce, Pig, Sqoop.
- **Data Engineering & Warehousing:** ETL/ELT Pipelines, Data Modeling (Star & Snowflake Schema), Data Warehousing, Batch & Stream Processing, dbt (models, tests, snapshots), Snowflake, Amazon Redshift, Google BigQuery, Amazon RDS.
- **Workflow Orchestration & DevOps:** Apache Airflow, CI/CD Pipelines, Terraform (Infrastructure as Code), GitHub, Cloud Monitoring (AWS CloudWatch).
- **Machine Learning:** Scikit-Learn, TensorFlow, PyTorch, Keras, Neural Networks (CNN, RNN, LSTM), Transformers, Hugging Face, Embeddings, Tokenization, LLM Fine-Tuning, RAG, GANs.
- **Visualization & Analytics:** Tableau, Looker, Google Sheets, Python (Matplotlib, Seaborn), Microsoft Excel (Advanced Excel, VLOOKUP, Solver, VBA), Google Analytics.
- **Productivity Tools:** Microsoft Office (Word, PowerPoint, Outlook, Access)

## PROJECTS EXPERIENCE

**Diabetes Readmission Analysis**, University of North Texas — Aug 2025 - Dec 2025

- Analyzed 100K+ hospital records using Python and SQL to identify key drivers of patient readmissions, including medication changes (35%), A1C levels, and hospital stay duration.
- Built and tuned Random Forest and Logistic Regression models, improving high-risk patient detection accuracy by 18% and enabling proactive care interventions and designed 5 interactive Tableau dashboards to monitor readmission trends.

**AI Waitress – Voice-Powered Restaurant Order Assistant**, University of North Texas — Aug 2025 - Dec 2025

- Designed and deployed an AI-powered voice assistant for restaurant order automation, reducing manual order handling by 60% and achieving 95% speech-to-intent recognition accuracy.
- Engineered a low-latency FastAPI-based multi-threaded system integrating Speech-to-Text, LLaMA-based LLM, and Text-to-Speech, delivering real-time responses with <1.5s average latency and seamless POS integration.